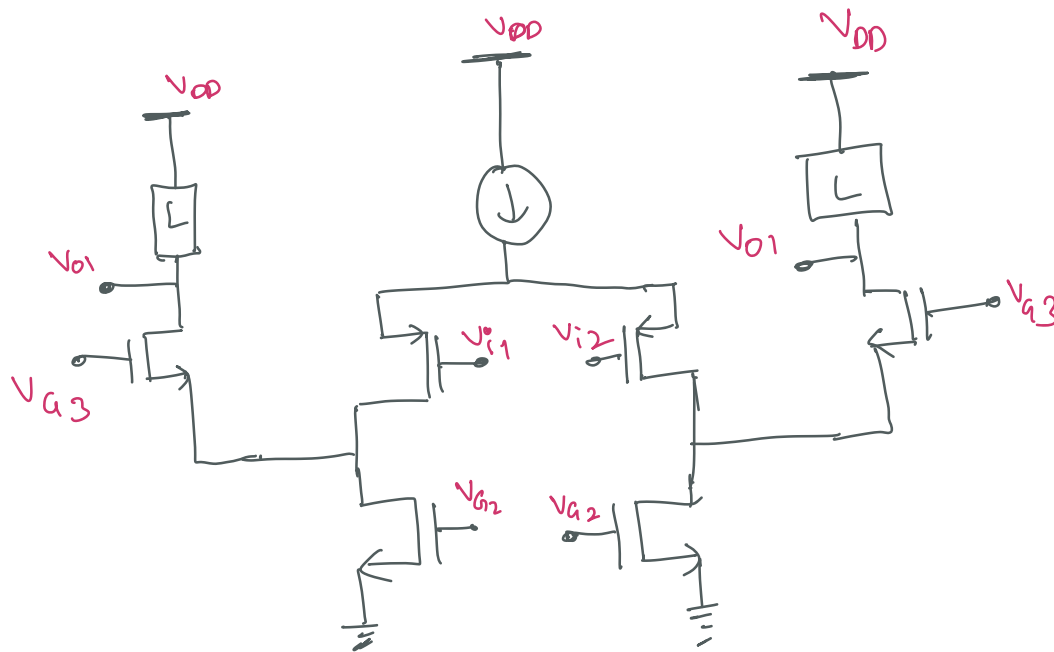


Lecture-14

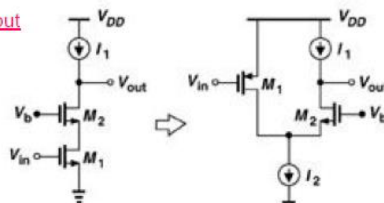
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Solution of Homework: Implementation of Differential
pair using PMOS Folded cascode
stage.

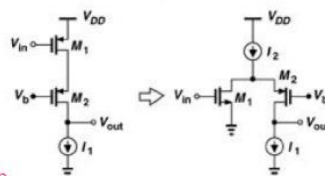


Folded Cascode Amplifier

PMOS Input



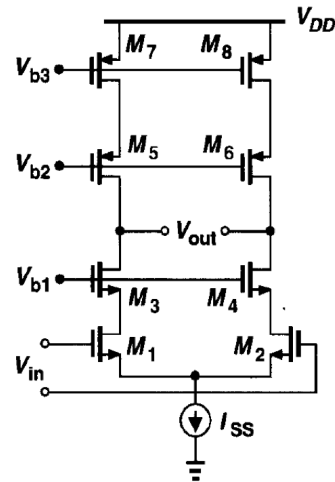
NMOS Input



- + High Gain
- + Output Swing Decoupled from Input Swing
- Reduced Speed

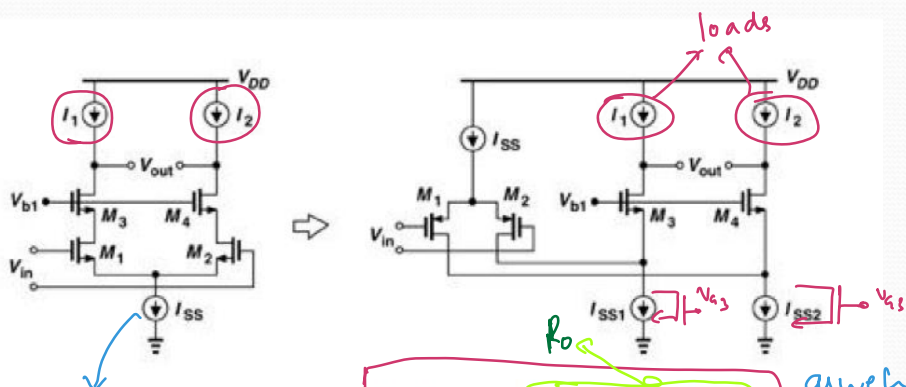
Telescopic Cascode Amplifier

Cascode...



63

Differential Folded Cascode Opamp Topology

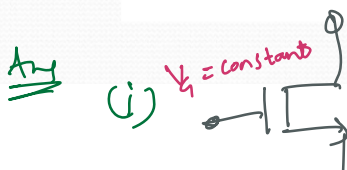


Q How do we Generate a constant current source?

$$|A_v| \approx g_{m1} [(g_{m3} + g_{mb3}) r_{o3}] r_{o1}$$

Cascode Gain

as we have drawn in the 1st page of the lecture.

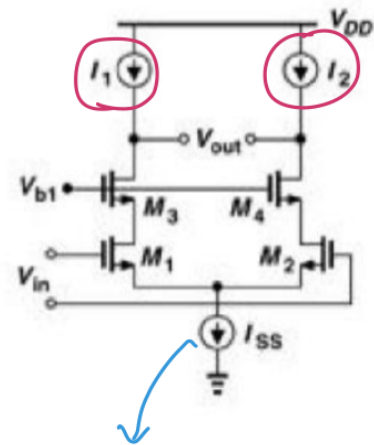


64



* $\Rightarrow$ There is another interesting modification which we can apply to these topologies above. This is related to the transformation of "Diff. op." that we have in the solution into a "single end op."

$\rightarrow$ One way to do that is to transform the Independent current sources which are acting like loads into Current mirrors



$\rightarrow$ But to understand how current mirrors are used in the Diff. Amplifier, we should first study Current Generators and Current mirrors.

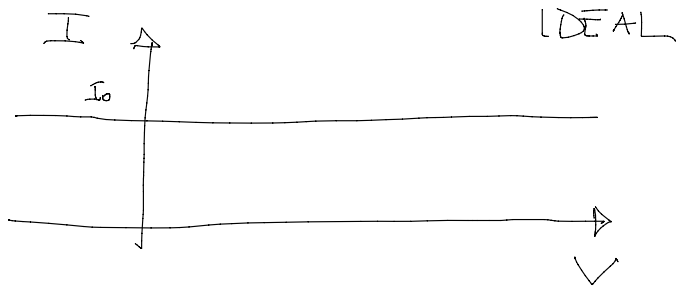


Current Mirrors

Current Generators



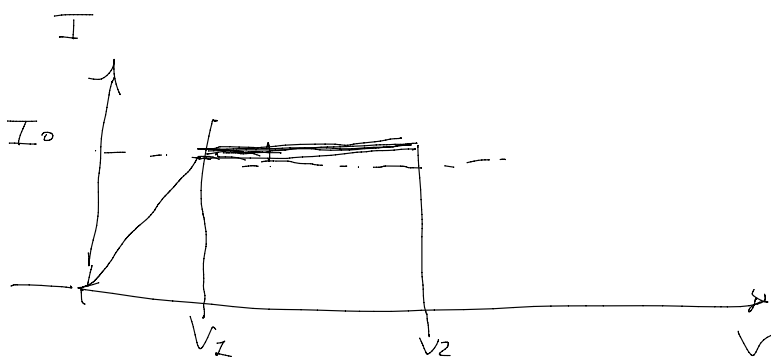
Current Generators



$$I = I_0 \quad \forall V$$

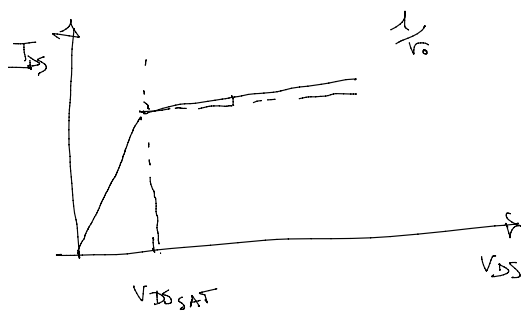
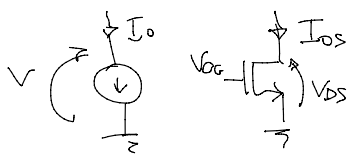
$$\frac{\partial I}{\partial V} = 0 = \frac{1}{r} \rightarrow r = \infty$$

REAL



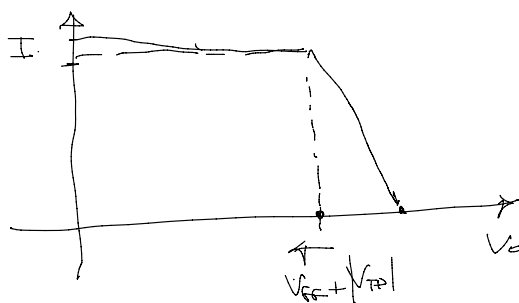
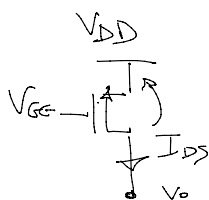
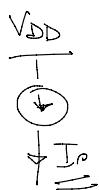
$$I \sim I_0 \quad \text{per } V_1 < V < V_2$$

$$r = k \neq \infty$$



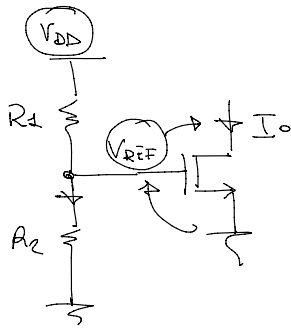
$$\frac{\partial I_{DS}}{\partial V_{DS}} = g_o = \lambda I_D = \frac{1}{r_o}$$

$$V > V_{ov}$$



$$r_o = \frac{1}{\lambda_p I_{DS}}$$

Ideality / Reality



$$S_X^Y = \lim_{\Delta X \rightarrow 0} \frac{\frac{\Delta Y}{Y}}{\frac{\Delta X}{X}} = \lim_{\Delta X \rightarrow 0} \frac{X}{Y} \cdot \frac{\Delta Y}{\Delta X} = \frac{X}{Y} \frac{\partial Y}{\partial X}$$

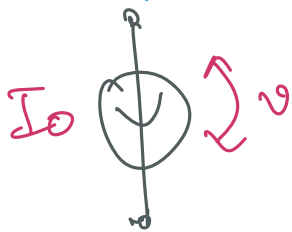
$$S_{V_{DD}}^{V_{REF}} = \frac{R_2}{R_1 + R_2} \cdot \frac{V_{DD}}{V_{REF}} = \frac{1}{1}$$

$$V_{REF} = V_{DD} \cdot \frac{R_2}{R_1 + R_2}$$

$$\frac{\partial V_{REF}}{\partial V_{DD}}$$

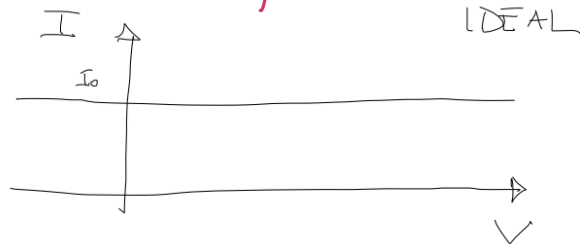
$$S_{V_{REF}}^{I_O} = \frac{\partial I_{DS}}{\partial V_{GS}} \cdot \frac{V_{REF}}{I_{DS}} = \frac{2I_O}{V_{REF} - V_T} \cdot \frac{V_{REF}}{I_O} > 2$$

* The ideal current Mirror would be a device which is characterised by the characteristics that it will Generate the same current no matter the inputs.



perfect current Generator

current generators
marched 24 marzo 2021 12:37



$$I = I_O \quad \forall V$$

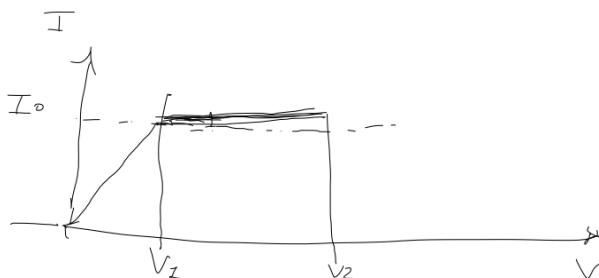
$$\left(\frac{\partial I}{\partial V} \right) = 0 = \frac{1}{r} \rightarrow r = \infty$$

Ideal
o/p Resistance of current
Generator is ∞

derivative of
the current
w.r.to voltage

→ Real world Current Generator

REAL



$$I \sim I_0$$

for

$$V_1 < V < V_2$$

$$r = k \neq \infty$$

The current is approximately constant

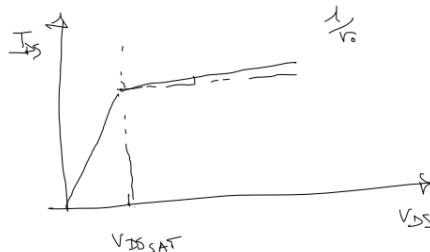
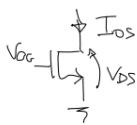
In a particular interval

Actual o/p resistance of Realistic current generator is 'r' which is High but not equal to infinity.

NMOS Transistor as a current Generator

(in saturation)

In saturation I_D is almost constant.



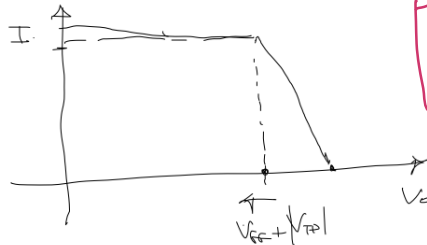
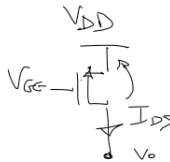
$$\frac{\partial I_{D0}}{\partial V_{DS}} = g_o = \lambda I_{D0} = \frac{1}{r_o}$$

$$V > V_{ov}$$

$$\frac{\partial I_D}{\partial V_{DS}} = g_D = \lambda I_D = \frac{1}{r_D}$$

$$r_D = \frac{1}{\lambda I_D} \rightarrow \text{o/p Resistance}$$

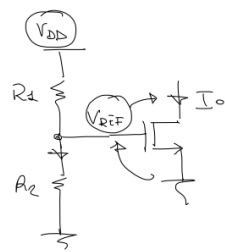
PMOS in saturation as current generator



$$r_D = \frac{1}{\lambda_p I_{DS}}$$

→ Constant struggle of Analog Designers is to balance b/w Ideality & Reality.

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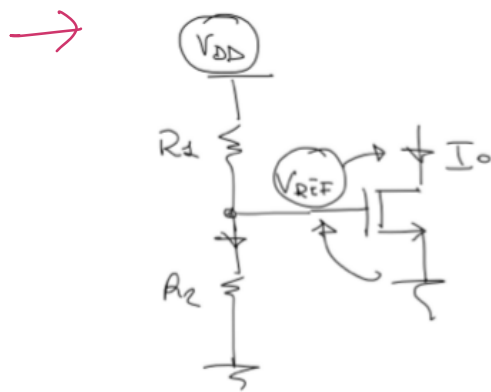
$$S_X^Y = \lim_{\Delta X \rightarrow 0} \frac{\frac{\Delta Y}{Y}}{\frac{\Delta X}{X}} = \lim_{\Delta X \rightarrow 0} \frac{X}{Y} \cdot \frac{\Delta Y}{\Delta X} = \frac{X}{Y} \frac{\partial Y}{\partial X}$$

$$S_{V_{DD}}^{V_{REF}} = \frac{R_2}{R_1 + R_2} \cdot \frac{V_{DD}}{V_{REF}} = \underline{\underline{1}}$$

$$V_{REF} = V_{DD} \cdot \frac{R_2}{R_1 + R_2}$$

$$\frac{\partial V_{REF}}{\partial V_{DD}}$$

$$S_{V_{REF}}^{I_D} = \frac{\partial I_{DS}}{\partial V_{GS}} \cdot \frac{V_{REF}}{I_{DS}} \approx \frac{2I_D}{V_{REF} - V_t} \cdot \frac{V_{REF}}{I_D} > 2$$



Using proper biasing techniques like voltage Divider Bias we can ^{bias} maintain constant voltage across V_{GS} and thereby constant current I_D .

→ But, This is not Ideal because the supply (V_{DD}) voltage is also prone to minute variations. Then current is also prone to some change.

How to characterize dependency of I_D w.r to supply voltage V_{DD} ?

Ans well we can apply sensitivity Analysis

S_X^Y = sensitivity of Y w.r to X

$$S_X^Y = \lim_{\Delta X \rightarrow 0} \frac{\frac{\Delta Y}{Y}}{\frac{\Delta X}{X}} = \lim_{\Delta X \rightarrow 0} \frac{X}{Y} \left(\frac{\Delta Y}{\Delta X} \right)$$

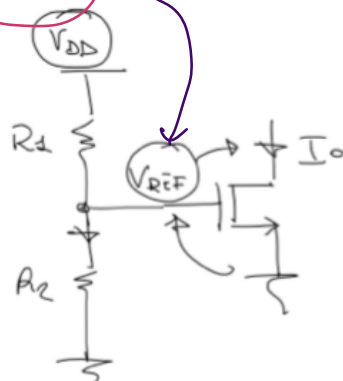
$$S_X^Y = \frac{X}{Y} \lim_{\Delta X \rightarrow 0} \frac{\Delta Y}{\Delta X}$$

$$S_X^Y = \frac{X}{Y} \frac{\partial Y}{\partial X}$$

For example:- we may be interested to know

How variations in V_{DD} will effect the

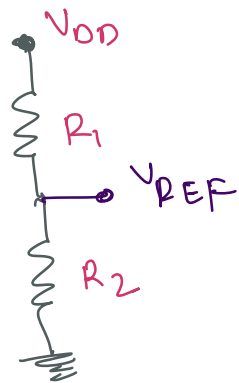
V_{REF} voltage which control V_{GS} of the Transistor



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$S_{V_{DD}}^{V_{REF}}$ = will tell in variation in V_{REF} w.r.to V_{DD} .

$$V_{REF} = V_{DD} \cdot \frac{R_2}{R_1 + R_2}$$



$$S_{V_{DD}}^{V_{REF}} = \frac{V_{DD}}{V_{REF}} \cdot \frac{\partial V_{REF}}{\partial V_{DD}}$$

$$\Rightarrow \frac{R_1 + R_2}{R_2} \cdot \frac{R_2}{R_1 + R_2}$$

$$S_{V_{DD}}^{V_{REF}} \Rightarrow 1$$

Sensitivity is equal to ONE.

→ let's study sensitivity of current I_O w.r.to V_{REF}

$$S_{V_{REF}}^{I_O} = \frac{V_{REF}}{I_O} \frac{\partial I_O}{\partial V_{REF}}$$

$$V_{REF} \equiv V_{GS}$$

$$\Rightarrow \frac{V_{GS}}{I_O} \frac{\partial I_O}{\partial V_{GS}}$$

$$\Rightarrow \frac{V_{GS}}{I_O} \cdot g_m$$

$$\Rightarrow \frac{V_{GS}}{I_O} \cdot \frac{2I_O}{V_{OV}}$$

$$S_{V_{REF}}^{I_O} \Rightarrow \frac{2V_{GS}}{V_{OV}} > 2$$

There is High sensitivity of current wrto variation in voltage.

∴ It's not so better solution to implement current Generator.

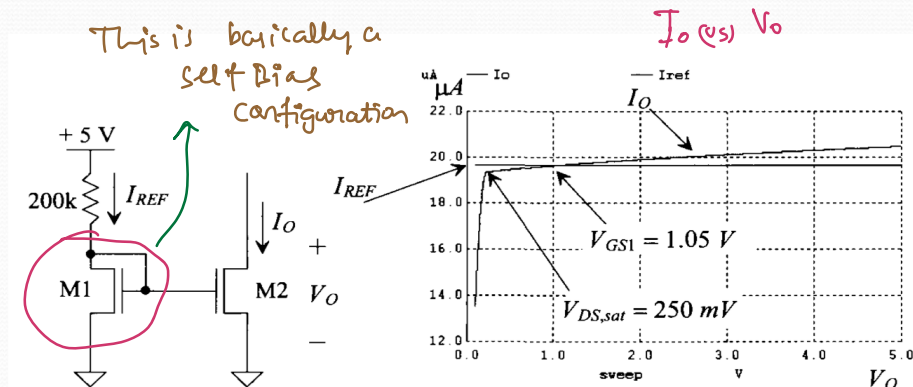
Voltage and Current References, Current Mirrors

References

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- EK:** M. Hekmat (Stanford): *Fundamentals o Analog Integrated Cirtcuis Design*, Handout, Summer 2010
- MA:** O. Oliaei (Univ. of Massachusets) *Analog Integrated Circuits Design Handout*, 2010
- GA:** B. Boser, A. Kiknejad, S. Gambini (Berkeley) *Advanced Analog Integrated Circuits*, Handout, Spring 2011
- YU:** F.Yuan (Yerson Univ. Toronto) *CMOS Circuits for Passive Wireless Microsystems*, 2011
- CA:** T. Carusone, D. Johns, K. Martins *Analog Integrated Circuits Design*, J. Wiley (2012)
- MU:** B.Murmann (Stanford) *Advanced Analog Integrated Circuits*, Handout, Spring 2013.
- SS:** E. Sanchez-Sinencio (Texas) *Advanced Analog Circuits Design Techniques*, Handout, Spring 2013
- BK:** J. Baker *CMOS Circuit Design Layout and Simulation*, 3rd edition



The basic current reference topology



→ In this case rather than partitioning supply voltage using voltage divider Bias configuration.

→ one of the Resistor is replaced by a Transistor

→ This configuration offers an improved constant current source.

Sensitivity

- Goal: We want the reference current to be insensitive to:
 - supply variations.
 - temperature variations.
 - process variations.

The sensitivity of parameter y to a change in parameter x is defined as:

$$S_x^y \triangleq \lim_{\Delta x \rightarrow 0} \frac{\frac{\Delta y}{y}}{\frac{\Delta x}{x}} = \frac{x}{y} \frac{\partial y}{\partial x}$$

In our case we are interested in sensitivity of the output current to supply voltage:

$$S_{V_{DD}}^{I_o} = \frac{V_{DD}}{I_o} \times \frac{\partial I_o}{\partial V_{DD}}$$

Example: for sensitivity of 1, a 10% change in supply results in a 10% change in the output current.

Current sensitivity

$$I_o \simeq \frac{\left(\frac{W}{L}\right)_2}{\left(\frac{W}{L}\right)_1} \times I_{Ref} = \frac{\left(\frac{W}{L}\right)_2}{\left(\frac{W}{L}\right)_1} \times \frac{V_{DD} - V_t - V_{OV}}{R}$$

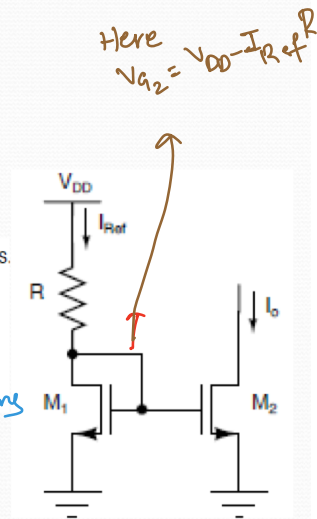
The current is directly proportional to V_{DD} ; thus it is sensitive to supply variations.

If V_{DD} is much larger than V_{GS} :

$$S_{V_{DD}}^{I_o} \simeq 1$$

This also much high for many applications

The circuit is too sensitive to supply for most applications!



4

MOS Widlar source

$$\sqrt{I_{OUT}} = \frac{-\sqrt{\frac{2}{k'(W/L)_2}} \pm \sqrt{\frac{2}{k'(W/L)_2} + 4R_2 V_{ov1}}}{2R_2}$$

Differentiating with respect to V_{DD} gives

$$\frac{1}{2\sqrt{I_{OUT}}} \frac{\partial I_{OUT}}{\partial V_{DD}} = \frac{1}{4R_2} \frac{1}{\sqrt{\frac{2}{k'(W/L)_2} + 4R_2 V_{ov1}}} 4R_2 \frac{\partial V_{ov1}}{\partial V_{DD}}$$

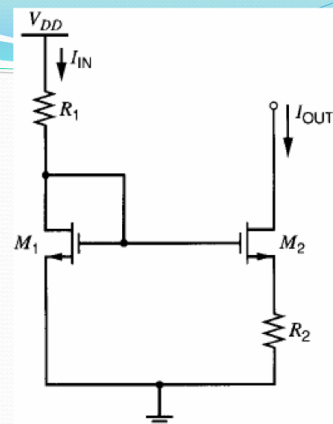
$$\frac{\partial V_{ov1}}{\partial V_{DD}} = \sqrt{\frac{2}{k'(W/L)_1}} \frac{1}{2\sqrt{I_{IN}}} \frac{\partial I_{IN}}{\partial V_{DD}} = \frac{V_{ov1}}{2I_{IN}} \frac{\partial I_{IN}}{\partial V_{DD}}$$

$$S_{V_{DD}}^{I_{OUT}} = \frac{V_{ov1}}{\sqrt{V_{ov2}^2 + 4I_{OUT}R_2V_{ov1}}} S_{V_{DD}}^{I_{IN}}$$

Since I_{OUT} is usually much less than I_{IN} , V_{ov2} is usually small and $I_{OUT}R_2 \approx V_{ov1}$

$$S_{V_{DD}}^{I_{OUT}} \approx \frac{V_{ov1}}{\sqrt{4V_{ov1}^2}} S_{V_{DD}}^{I_{IN}} = 0.5 S_{V_{DD}}^{I_{IN}} \approx 1$$

Thus for this case, a 10 percent power-supply voltage change results in a 5 percent change in I_{OUT}



GM

5

MOS Widlar source

$$\sqrt{I_{OUT}} = \frac{-\sqrt{\frac{2}{k'(W/L)_2}} + \sqrt{\frac{2}{k'(W/L)_2} + 4R_2V_{ov1}}}{2R_2}$$

Differentiating with respect to V_{DD} gives

$$\frac{1}{2\sqrt{I_{OUT}}} \frac{\partial I_{OUT}}{\partial V_{DD}} = \frac{1}{4R_2} \frac{1}{\sqrt{\frac{2}{k'(W/L)_2} + 4R_2V_{ov1}}} 4R_2 \frac{\partial V_{ov1}}{\partial V_{DD}}$$

$$\frac{\partial V_{ov1}}{\partial V_{DD}} = \sqrt{\frac{2}{k'(W/L)_1}} \frac{1}{2\sqrt{I_{IN}}} \frac{\partial I_{IN}}{\partial V_{DD}} = \frac{V_{ov1}}{2I_{IN}} \frac{\partial I_{IN}}{\partial V_{DD}}$$

$$S_{V_{DD}}^{I_{OUT}} = \frac{V_{ov1}}{\sqrt{V_{ov2}^2 + 4I_{OUT}R_2V_{ov1}}} S_{V_{DD}}^{I_{IN}}$$

Since I_{OUT} is usually much less than I_{IN} , V_{ov2} is usually small and $I_{OUT}R_2 \approx V_{ov1}$

$$S_{V_{DD}}^{I_{OUT}} = \frac{V_{ov1}}{\sqrt{4V_{ov1}^2}} S_{V_{DD}}^{I_{IN}} = 0.5 S_{V_{DD}}^{I_{IN}} \approx 1$$

$$V_{ov1} = \sqrt{\frac{2I_{IN}}{\beta_1}}$$

Thus for this case, a 10 percent power-supply

voltage change results in a 5 percent change in I_{OUT}

$$\frac{\partial V_{ov1}}{\partial I_{IN}}$$

5

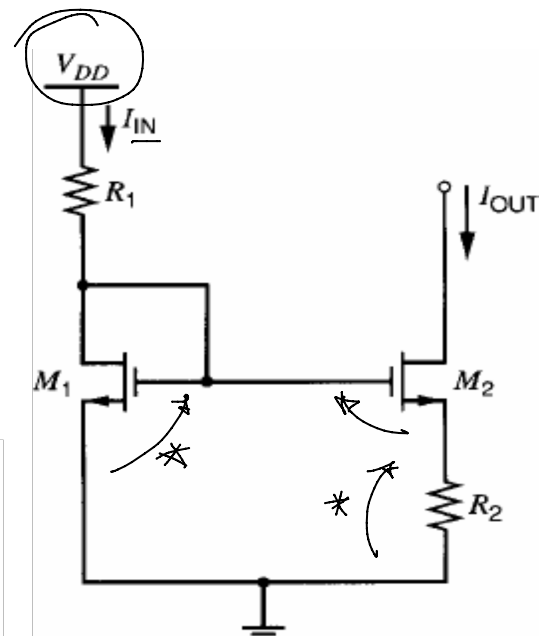
$$I_D = \frac{\beta}{2} V_{ov}^2$$

$$\beta = \beta' \frac{W}{L}$$

$$V_{ov} = \sqrt{\frac{2I_D}{\beta}}$$

$$V_{TN} + V_{ov1} \neq V_{ov2} + V_{TN} + I \cdot R_2$$

$$I_0 \cdot R_2 + \sqrt{\frac{2I_0}{\beta_2}} - V_{ov1} = 0 \quad \text{for } \sqrt{I_0}$$



$$\left[I_o \cdot R_2 + \sqrt{\frac{2I_o}{\beta_2}} - V_{ov1} = 0 \right] \quad \text{for } \sqrt{I_o}$$

$$\sqrt{I_o} = \frac{-\sqrt{\frac{2}{\beta_2}} + \sqrt{\frac{2}{\beta_2} + 4R_2 \cdot V_{ov1}}}{2 \cdot R_2}$$

$$\left(\frac{\partial I_o}{\partial V_{DD}} \right) = \frac{\frac{\partial \sqrt{I_o}}{\partial V_{DD}} \cdot \frac{\partial I_o}{\partial \sqrt{I_o}}}{\frac{\partial V_{ov1}}{\partial V_{DD}} \cdot \frac{\partial V_{ov1}}{\partial I_o} \cdot \frac{\partial I_o}{\partial \sqrt{I_o}}} \quad \begin{matrix} 2^{nd} & 3^{rd} & 1^{st} \\ \downarrow & \downarrow & \downarrow \end{matrix}$$

$$\frac{\partial \sqrt{I_o}}{\partial I_o} = \frac{1}{2} \cdot \frac{1}{\sqrt{I_o}} \Rightarrow \frac{\partial I_o}{\partial \sqrt{I_o}} = 2 \sqrt{I_o} \quad 1^{st}$$

$$\frac{\partial \sqrt{I_o}}{\partial V_{ov1}} = \frac{1}{\cancel{2\beta_2}} \cdot \left(\frac{1}{\cancel{2}} \cdot \frac{1}{\sqrt{\frac{2}{\beta_2} + 4R_2 \cdot V_{ov1}}} \right) \cdot \cancel{1 \cdot R_2} \quad 2^{nd}$$

$$\frac{\partial V_{ov1}}{\partial V_{DD}} = \frac{\partial V_{ov1}}{\partial I_{IN}} \cdot \frac{\partial I_{IN}}{\partial V_{DD}} \quad 3^{rd}$$

$$V_{ov1} = \sqrt{\frac{2I_{IN}}{\beta_1}}$$

$$\frac{\partial V_{ov1}}{\partial I_{IN}} = \sqrt{\frac{2}{\beta_1}} \cdot \frac{1}{2\sqrt{I_{IN}}} \cdot \frac{\sqrt{I_{IN}}}{\sqrt{I_{IN}}} = \frac{1}{2} \frac{V_{ov1}}{I_{IN}}$$

$$\partial I_o$$

$$\left[\frac{\partial \sqrt{I_0}}{\partial V_{ov1}} \cdot \frac{\partial I_0}{\partial \sqrt{I_0}} \cdot \frac{\partial V_{ov1}}{\partial V_{DD}} \right] \cdot \frac{V_{DD}}{I_0} = S_{V_{DD}}^{I_0} \approx \frac{\partial I_0}{\partial V_{DD}}$$

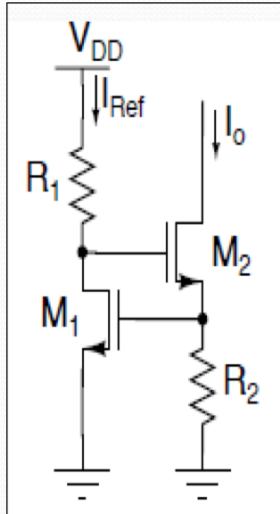
$$= \frac{1}{\sqrt{\frac{2}{\beta_2} + 4R_2 V_{ov1}}} \cdot 2\sqrt{I_0} - \frac{1}{2} \frac{V_{ov1}}{I_{IN}} \cdot \frac{\partial I_{IN}}{\partial V_{DD}} \cdot \frac{V_{DD}}{I_0} = S_{V_{DD}}^{I_{IN}}$$

$$= \frac{V_{ov1}}{\sqrt{\frac{2}{\beta_2} + 4R_2 V_{ov1}}} \cdot \frac{I_{IN}}{\sqrt{I_0}} = S_{V_{DD}}^{I_{IN}}$$

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Sources Using V_t Standards



Can reduce sensitivity by making I_o depend on voltages other than V_{DD} , e.g., V_t .

$$I_o = \frac{V_{GS1}}{R_2} = \frac{V_t + V_{OV1}}{R_2} = \frac{V_t + \sqrt{\frac{2I_{Ref}}{\mu C_{ox}(\frac{W}{L})_1}}}{R_2}$$

If I_{Ref} is small and $(W/L)_1$ is sufficiently large $V_{OV} \ll V_t$:

$$I_o \simeq \frac{V_t}{R_2}$$

Therefore, this circuit is known as a *threshold-referenced* bias circuit.

$$\begin{aligned} S_{V_{DD}}^{I_o} &= S_{I_{Ref}}^{I_o} \times S_{V_{DD}}^{I_{Ref}} \\ &\simeq \frac{I_{Ref}}{I_o} \times \frac{\partial I_o}{\partial I_{Ref}} \times \underbrace{S_{V_{DD}}^{I_{Ref}}}_{\simeq 1} \\ &\simeq \frac{1}{2} \frac{V_{OV1}}{V_t + V_{OV1}} \end{aligned}$$

For example, if $V_t = 1$ V, $V_{ov1} = 0.1$ V, and $S_{V_{DD}}^{I_{IN}} \simeq 1$

$$S_{V_{DD}}^{I_{OUT}} \simeq \frac{0.1}{2(1.1)} \simeq 0.045$$

by

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